Agent_2

https://gemini.google.com/app/532c89661d8aa503

User prompt: you are part of a big mission and your task is given in the md file

Response: 1. Decoupled Horizontal ODE

The horizontal motion of the particle is governed by the 1D time-independent Schrödinger equation for a free particle along the x-axis:

$$-\frac{\hbar^2}{2m}\frac{d^2X(x)}{dx^2} = E_x X(x) \quad [\text{cite: 1}]$$

Rearranging into standard second-order differential form gives:

$$\frac{d^2X(x)}{dx^2} + k_x^2 X(x) = 0$$

where $k_x \equiv \frac{2mE_x}{\hbar}$ is the horizontal wavenumber.

2. Spatial Wave Solution $X(x)$

The general solution to this differential equation in complex exponential form is:

$$X(x) = Ae^{ik_x x} + Be^{-ik_x x}$$

For a projectile launched forward in the positive x-direction with well-defined positive horizontal momentum $p_x > 0$, set $B = 0$:

$$X(x) = Ae^{ik_x x}$$

3. Derivation of the Horizontal Wavevector k_x

Classically, a particle launched with initial speed u at angle θ relative to the horizontal ground has a horizontal velocity component:

$$v_x = u\cos\theta$$

The kinetic energy associated with horizontal motion is:

$$E_x = \frac{1}{2}mv_x^2 = \frac{1}{2}mu^2\cos^2\theta$$

Substituting E_x into the expression for k_x:

$$k_x = \frac{2m\left(\frac{1}{2}mu^2\cos^2\theta\right)}{\hbar} = \frac{m^2u^2\cos^2\theta}{\hbar} = \frac{mu\cos\theta}{\hbar}$$

This confirms consistency with the de Broglie relation $p_x = \hbar k_x = mu\cos\theta$.

4. 1D Spatial Probability Density Normalization Integral

Because $X(x)$ represents an unbounded free-particle state along the horizontal direction, normalization is defined using standard continuum delta-normalization or box normalization.

Dirac Delta Normalization (Continuous Spectrum)

The inner product normalization condition over the spatial domain $(-\infty, \infty)$ is set up as:

$$\int_{-\infty}^{\infty} X_{k_x'}^*(x) X_{k_x}(x)\, dx = |A|^2 \int_{-\infty}^{\infty} e^{i(k_x - k_x')x}\, dx = |A|^2 \cdot 2\pi\delta(k_x - k_x') = \delta(k_x - k_x')$$

Solving for the normalization constant A:

$$|A| = \frac{1}{2\pi} \implies X(x) = \frac{1}{2\pi} e^{ik_x x}$$

Box Normalization (Finite Length L)

Restricting the horizontal domain to a normalization length L:

$$\int_0^L |X(x)|^2\, dx = |A|^2 \int_0^L dx = |A|^2 L = 1 \implies A = \frac{1}{L}$$

$$X(x) = \frac{1}{L} e^{i\left(\frac{mu\cos\theta}{\hbar}\right)x}$$
